KUET · ESE 1205 · Thermodynamics for Energy Engineering

ESE 1205 Question Map

Every question from the 2017, 2018, 2019, 2020, 2021, 2022, 2023 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 1205 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 5 h for Part A and 4 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×7, 2022 ×6, 2021 ×5, 2020 ×4, 2019 ×3, 2018 ×2, 2017 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer for each block first (30 minutes), then follow this list. Psychrometry (B8) and gas mixtures (B7) are the biggest mark-earners in Part B and both have calculators; the Otto/Diesel numerical (B4) comes every year. In Part A the first law (A5, A6) and the second law (A7) are the safest. Entropy and exergy (A8, A9) are harder: do them last, using the calculators to practise.

Section A: Concepts, first law, second law, entropy, exergy

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
A5 Zeroth and first law; energy balance for closed and open systems; boundary work	Very likely 100%	Medium	13%
A7 Second law: statements, heat engines, reversibility, Carnot principles and proofs	Very likely 100%	Medium	24%

A6 First-law numericals for steady-flow devices (turbine, compressor, pump, heat exchanger)	Very likely 86%	Medium	37%
A3 Pure substance, phase rule, state postulate, T-v diagram, quality	Very likely 82%	Medium	45%

Block 3: very likely numericals · 2.0 h

Topic	Chance	Level	Covered
A9 Exergy: dead state, closed-system and flow exergy, exergy balance, second-law efficiency	Very likely 100%	Hard	59%
A8 Entropy: Clausius inequality, Tds equations, entropy balance, isentropic efficiency	Very likely 100%	Hard	72%

Block 4: likely · 1.6 h

Topic	Chance	Level	Covered
A1 Basic concepts: system, viewpoints, properties, equilibrium, process	Likely 79%	Easy	79%
✔ Pass line: 79% of Section A marks covered, enough to choose 3 of the 4 questions (about 5.4 h)			
A2 Energy: energy mind map, stored vs transient energy, heat vs work, path and point functions	Likely 75%	Easy	86%
A4 Ideal gas and real gas: equations of state, compressibility factor	Likely 75%	Medium	96%

Only if time is left · 1.0 h

Topic	Chance	Level	Covered
A10 Thermodynamic relations: Maxwell relations, expansivity, compressibility, Joule-Thomson	Possible 46%	Hard	100%

Section B: Power cycles, refrigeration, gas mixtures, psychrometry

Block 1: very likely and easy · 0.8 h

Topic	Chance	Level	Covered
B6 Refrigeration: vapour compression and vapour absorption systems	Very likely 100%	Easy	9%
B1 Power cycle basics: air-standard assumptions, gas vs vapour cycles, Carnot cycle	Very likely 100%	Easy	17%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
B8 Psychrometry: moist-air properties, derivations, chart and numericals	Very likely 100%	Medium	38%

B7 Ideal gas mixtures: Dalton, Amagat, mole and mass fractions, mixture properties	Very likely 100%	Medium	57%
B4 Otto, Diesel and Dual cycles (IC engines): diagrams, derivations, comparisons, numericals	Very likely 100%	Medium	73%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B3 Rankine cycle numericals (simple, actual, reheat, regenerative)	Very likely 96%	Hard	85%

✔ Pass line: 85% of Section B marks covered, enough to choose 3 of the 4 questions (about 4.1 h)

Block 4: likely · 1.5 h

Topic	Chance	Level	Covered
B2 Rankine cycle theory: ideal vs actual, efficiency improvement, superheat, reheat, regeneration	Likely 79%	Medium	95%
B5 Brayton (gas turbine) cycle	Likely 64%	Medium	100%

Pass plan

Seven papers (2017-2023). Normal papers: 210 marks, answer any THREE of four questions in each section (35 marks per question). 2020 was a short exam (120 marks, any TWO of three per section). Section A (Q1-Q4) = basic concepts, pure substance and gases, first law, second law, entropy, exergy, relations. Section B (Q5-Q8) = power cycles (Rankine, Otto, Diesel, Dual, Brayton), refrigeration, gas mixtures and psychrometry. **Minimum pass set:** the "Pass set" topics let you write three full questions in each section: in Section A, Q1 (concepts, pure substance), Q2 (first law + device numerical) and Q3 (second law + entropy); in Section B, one cycle question (Rankine or IC engine), Q7 (gas mixture) and Q8 (psychrometry). These are the most predictable questions on the paper and their numericals repeat almost word for word. Answers follow the course slides (Tahir Mahmud, Afsar/Sohag lectures, Cengel and Moran-Shapiro examples used in the slides); steam values are Cengel/Moran table values, checked with an IAPWS steam-property program.

SECTION A: CONCEPTS, FIRST LAW, SECOND LAW, ENTROPY, EXERGY

SECTION B: POWER CYCLES, REFRIGERATION, GAS

7 topics

cover 73% of all Section A marks asked

- Q1: system definitions, pure substance, T-v diagram, phase rule, ideal gas
- Q2: energy forms, first law, control-volume energy balance, device numericals
- Q3: Kelvin-Planck/Clausius, Carnot proofs, Tds, entropy balance
- Backup Q4: exergy derivation + numerical

MIXTURES, PSYCHROMETRY

5 topics

cover 78% of all Section B marks asked

- Rankine theory + one Rankine numerical (simple, reheat or regenerative)
- Otto/Diesel/Dual derivations and numericals
- Gas mixture numerical (mole fraction → M → R → V, ρ, c_p)
- Psychrometry: Carrier's equation numerical + definitions

- **Section A, pick Q1, Q2, Q3:** Q1 = definitions + pure substance/ideal gas numerical (A1, A3, A4); Q2 = first law statement + energy balance + a device numerical (A2, A5, A6); Q3 = second law statements + Carnot proofs + entropy (A7, A8). Q4 (exergy, A9) is the backup.
- **Section B, pick a cycle question, Q7 and Q8:** Rankine (B2 + B3) or IC engine cycles (B4); gas mixture numerical (B7, the same N_2 + CO_2 problem appears in 2019, 2020, 2022, 2023); psychrometry with Carrier's equation (B8, every year).
- **Repeated numericals (learn once, score many times):** compressor 1 bar/290 K → 7 bar/450 K (2017, 2022); turbine-driven pump (2017, 2021); reheat 15 MPa/600 °C/10.4 % moisture (2021, 2023); open FWH 15 MPa/1.2 MPa (2018, 2022); N_2 at 175 K (2021, 2022).
- **Derivations that repeat:** Tds equations (5 papers), closed-system exergy (5 papers), Carnot proofs (5 papers), Diesel efficiency, degree of saturation / relative humidity.
- **Always draw:** T-v diagram, energy mind map, p-v and T-s of each cycle, Rankine schematic + T-s, VCR/VAR schematic, psychrometric chart sketch.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: CONCEPTS, FIRST LAW, SECOND LAW, ENTROPY, EXERGY: MARKS ASKED PE

Topic	2017	2018	2019	2020	2021	2022	2023	Years	Ma
A5 Zeroth and first law; energy balance for closed and open systems; boundary work	6	21	27	24	10	25	10	7/7	1
A7 Second law: statements, heat engines, reversibility, Carnot principles and proofs	30	14	15	10	5	22	17	7/7	1
A6 First-law numericals for steady-flow devices (turbine, compressor, pump, heat exchanger)	29	30	15		20	10	15	6/7	1
A3 Pure substance, phase rule, state postulate, T-v diagram, quality	7	7	23	12		4	26	6/7	7
A9 Exergy: dead state, closed-system and flow exergy, exergy balance, second-law efficiency	10	12	15	20	37	22	15	7/7	1
A8 Entropy: Clausius inequality, Tds equations, entropy balance, isentropic efficiency	29	23	10	6	15	26	18	7/7	1
A1 Basic concepts: system, viewpoints, properties, equilibrium, process	6	12	23	6	10		10	6/7	6
A2 Energy: energy mind map, stored vs transient energy, heat vs work, path and point functions	10	10	12	12	20	10		6/7	7
A4 Ideal gas and real gas: equations of state, compressibility factor	24	11			23	21	9	5/7	8
A10 Thermodynamic relations: Maxwell relations, expansivity, compressibility, Joule-Thomson						23	20	2/7	4

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: POWER CYCLES, REFRIGERATION, GAS MIXTURES, PSYCHROMETRY: MARK

Topic	2017	2018	2019	2020	2021	2022	2023	Years	Ma
B6 Refrigeration: vapour compression and vapour absorption systems	18	12	10	10	12	12	8	7/7	8
B1 Power cycle basics: air-standard assumptions, gas vs vapour cycles, Carnot cycle	4	12	16	12	8	8	10	7/7	7
B8 Psychrometry: moist-air properties, derivations, chart and numericals	35	35	35	6	35	12	25	7/7	1
B7 Ideal gas mixtures: Dalton, Amagat, mole and mass fractions, mixture properties	30	35	25	24	23	25	10	7/7	1
B4 Otto, Diesel and Dual cycles (IC engines): diagrams, derivations, comparisons, numericals	24	28	27	18	15	10	27	7/7	1
B3 Rankine cycle numericals (simple, actual, reheat, regenerative)		18	16	20	15	18	13	6/7	1
B2 Rankine cycle theory: ideal vs actual, efficiency improvement, superheat, reheat, regeneration	15		11		20	17	32	5/7	9
B5 Brayton (gas turbine) cycle					12	15	15	3/7	4

Section A: Concepts, first law, second law, entropy, exergy

questions 1-4

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A5 Zeroth and first law; energy balance for closed and open systems; boundary work

Pass set · Very likely · 100% · Medium

Asked in 7 of 7 papers · 123 marks in total

STUDY THESE

- Zeroth law → temperature; first law → internal energy; second law → entropy
- First law for a cycle: $\oint \delta Q = \oint \delta W$; closed system: $Q - W = \Delta U$ (+ $\Delta KE + \Delta PE$)
- Control volume mass and energy rate balance; derivation for one inlet and one exit; steady-flow energy equation (SFEE)
- Boundary work $W = \int p\, dV$: constant pressure $p(V_2 - V_1)$, polytropic $(p_2V_2 - p_1V_1)/(1-n)$, isothermal $p_1V_1 \ln(V_2/V_1)$

2023 Q2(b) 10 marks

Write down the energy balance equation for a general steady-flow system and describe each term.

2022 Q1(a) 10 marks

With a proper example state the zeroth law of thermodynamics. Which property is based on this law?

2022 Q2(b) 15 marks

State the first law. For an open system with one inlet and one outlet, derive the mathematical form of the first law, considering no interactions other than mass flows and the associated volume-change work.

2021 Q2(a) 10 marks

State the zeroth and first laws. Write down the mass and energy rate balance equations for a control volume and mention what each term means.

2020 Q2(b) 9 marks

Which thermodynamic properties are based on the zeroth, first and second laws? Write the mathematical forms of the first law for a control volume and a control mass.

2020 Q2(c) 15 marks

A stationary gas is compressed without friction from 0.3 m^3, 0.105 MPa to 0.15 m^3, 0.105 MPa, while 37.6 kJ of heat leaves the gas. Calculate the change of internal energy.

2019 Q2(a) 5 marks

State the zeroth law of thermodynamics. Which thermodynamic property is based on this law?

> 2019 Q3(a) 10 marks
>
> State the first law of thermodynamics. Write down the energy rate equation for an open system and the meaning of each term.

> 2019 Q2(b) 12 marks
>
> 8 kg of gas expands in a flexible container with $Pv^{1.2}$ = constant from 1000 kPa, 1 m^3 to 5 kPa. If the specific internal energy decreases by 40 kJ/kg, find the heat transfer (magnitude and direction).

> 2018 Q2(b) 8 marks
>
> State the first law of thermodynamics. Write down the energy rate balance equation for a control volume and describe each term.

> 2018 Q1(d) 13 marks
>
> A gas in a piston-cylinder expands with pV^n = constant from p_1 = 3 bar, V_1 = 0.1 m^3 to V_2 = 0.2 m^3. Find the work for (i) n = 1.5 and (ii) n = 1.

> 2017 Q2(a) 6 marks
>
> State the first law of thermodynamics. Write down the energy rate balance for a control volume and mention what each term means.

Full answer: ESE 1205 Study Pack, topic A5.

A7 Second law: statements, heat engines, reversibility, Carnot principles and proofs

Pass set · Very likely · 100% · Medium

Asked in 7 of 7 papers · 113 marks in total

STUDY THESE

- Thermal reservoir; heat engine and its characteristics; thermal efficiency $\eta = W/Q_H = 1 - Q_L/Q_H$
- Kelvin-Planck and Clausius statements and their equivalence (violating one violates the other); 100 % efficiency is impossible
- Reversible vs irreversible processes; irreversibilities (friction, finite ΔT heat transfer, unrestrained expansion, mixing)

- Carnot corollaries: $\eta_{irr} < \eta_{rev}$; all reversible engines between the same reservoirs have the same η; $(Q_L/Q_H)_{rev} = T_L/T_H$; $\eta_{Carnot} = 1 - T_L/T_H$

> `2023` `Q3(b)` `10 marks`
>
> "The efficiencies of all reversible heat engines operating between the same two reservoirs are the same": prove it.

> `2023` `Q3(a)` `7 marks`
>
> Write down the characteristics of a heat engine and discuss its thermal efficiency.

> `2022` `Q3(a)` `10 marks`
>
> State the Clausius and Kelvin-Planck statements. With a schematic, explain their equivalence.

> `2022` `Q3(b)` `8 marks`
>
> Prove that an irreversible heat engine is always less efficient than a reversible one between the same two reservoirs.

> `2022` `Q4(b)` `4 marks`
>
> In the absence of friction and other irreversibilities, can a heat engine have 100 % efficiency? Explain.

> `2021` `Q3(a)` `5 marks`
>
> State the Clausius, Kelvin-Planck and entropy statements of the second law.

> `2020` `Q3(a)` `10 marks`
>
> "The efficiency of an irreversible heat engine is always less than that of a reversible one operating between the same two reservoirs": prove it.

> `2019` `Q4(a)` `7 marks`
>
> State the Kelvin-Planck statement of the second law and discuss its mathematical form.

> `2019` `Q4(d)` `8 marks`
>
> Short notes: (i) Carnot efficiency, (ii) irreversible process, (iii) Clausius inequality, (iv) thermal reservoir.

> `2018` `Q3(b)` `4 marks`
>
> Write down the Kelvin-Planck statement of the second law. (Tds part in A8.)

> `2018` `Q4(d)` `10 marks`
>
> Prove that $(Q_C/Q_H)_{rev\ cycle} = T_C/T_H$ (absolute temperatures).

> `2017` `Q3(a)` `6 marks`

State the Kelvin-Planck statement of the second law together with the Carnot corollaries.

2017 Q4(a) 17 marks

Prove that $(Q_L/Q_H)_{rev} = T_L/T_H$ with appropriate assumptions.

2017 Q5(b) 7 marks

Prove that the efficiency of a Carnot engine between T_1 and T_2 is $(T_1 - T_2)/T_1$.

Full answer: ESE 1205 Study Pack, topic A7.

A6 First-law numericals for steady-flow devices (turbine, compressor, pump, heat exchanger)

Pass set · Very likely · 86% · Medium

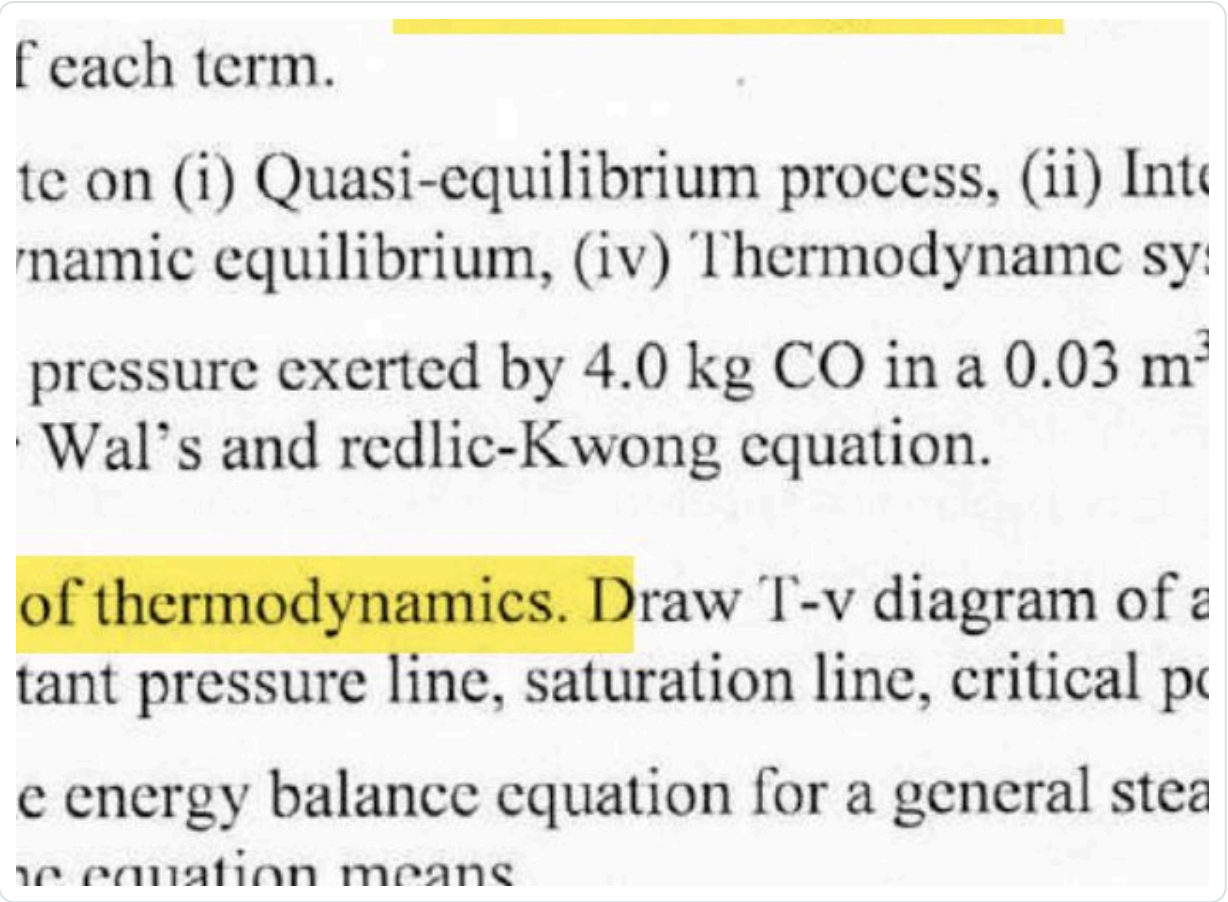
Asked in 6 of 7 papers · 119 marks in total

STUDY THESE

- SFEE per device: turbine $w = h_1 - h_2$ (+ΔKE, ΔPE), compressor/pump, heat exchanger energy balance, mixing
- Mass flow from volume flow: $m = AV/v = (AV)p/RT$ for an ideal gas
- Steam table look-ups: superheated, saturated mixture ($h = h_f + x\,h_{fg}$), compressed liquid ≈ $h_f(T)$
- Sign convention: Q into system +, W out of system +

2023 Q2(c) 15 marks

An adiabatic steam turbine produces 5 MW. Inlet 2 MPa, 400 °C, 50 m/s, 10 m; exit 15 kPa, x = 90 %, 180 m/s, 6 m. (i) Compare Δh, ΔKE, ΔPE; (ii) work per kg; (iii) mass flow.



`2022` `Q2(c)` `10 marks`

Same compressor (1 bar, 290 K, 6 m/s, 0.1 m^2 → 7 bar, 450 K, 2 m/s; heat loss 180 kJ/min): find the power input, and again neglecting the kinetic energy change (1 bar = 100 kPa).

`2021` `Q2(c)` `20 marks`

Same turbine-driven pump (steam 30 bar, 400 °C → 5 bar, 180 °C; water saturated liquid at 1 bar, 50 kg/s; h_2 = 417.69 kJ/kg; Q = −2 kW; 25 m). Find the pump power and the steam mass flow.

`2019` `Q3(c)` `15 marks`

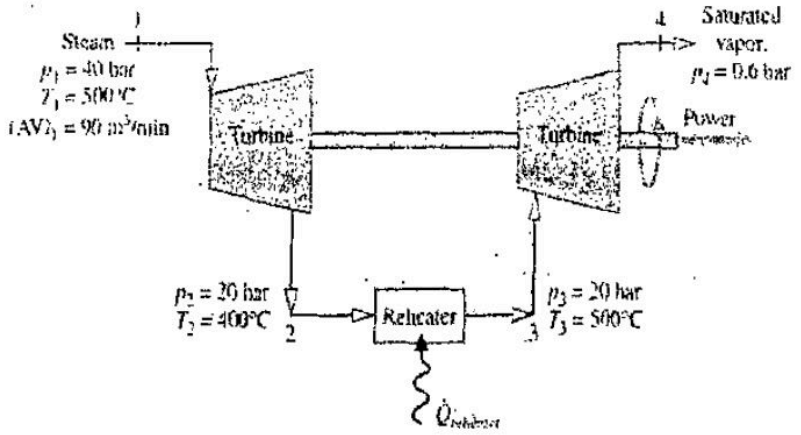
Waste heat recovery: water enters the steam generator at 290 kPa, 104 °C and leaves at 276 kPa, 160 °C, then expands in a turbine to 7 kPa, x = 90 %. Oven exhaust air enters at 182 °C, 100 kPa, 2 m^3/s and leaves at 138 °C. Power is worth Tk 7/kWh. Do you recommend the system? Support with calculations.

`2018` `Q2(c)` `15 marks`

Steam enters a first-stage turbine at 40 bar, 500 °C, 90 m^3/min, exits at 20 bar, 400 °C, is reheated at constant pressure to 500 °C and leaves the second stage as saturated vapour at 0.6 bar. Find (i) mass flow, (ii) total power, (iii) reheater heat rate.

what each term in the equation mean.

the first --stage turbine shown in figure at 40 *bar* and 500°C with volu
/min.

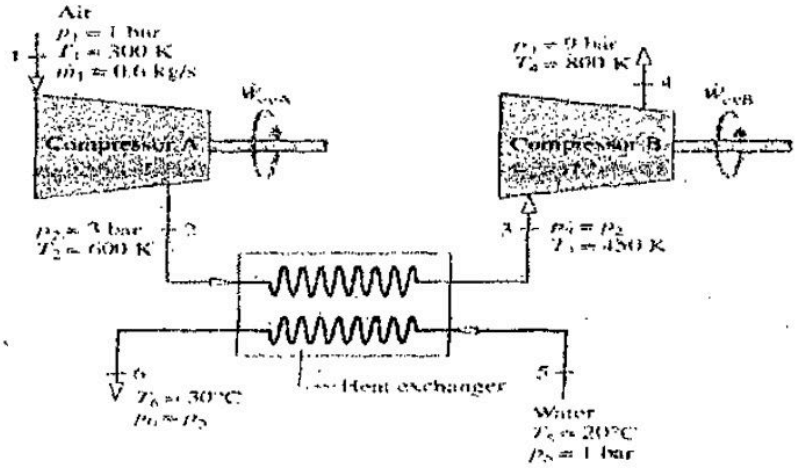


the turbine at 20 *bar* and 400°C .The steam is reheated at constant

2018 Q3(a) 15 marks

Air (1 bar, 300 K, 0.6 kg/s) is compressed in compressor A to 3 bar, 600 K, cooled in a heat exchanger to 450 K, then compressed in B to 9 bar, 800 K. Cooling water enters at 20 °C, 1 bar and leaves at 30 °C. c_p = 1.063 kJ/kg K. Find (i) total compressor power, (ii) water mass flow.

lwing figure.

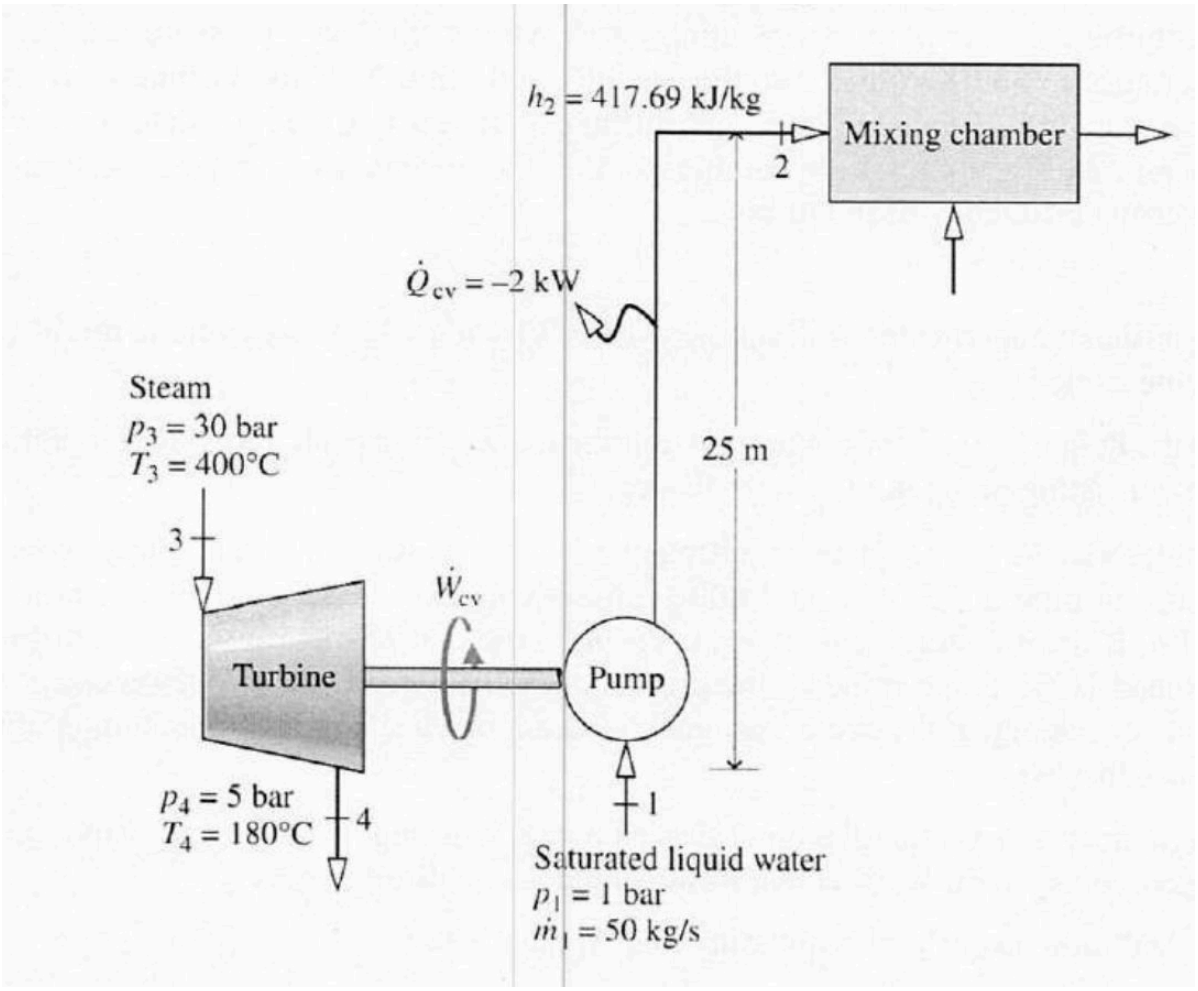


2017 Q2(b) 12 marks

Define a turbine and the kinds used in power generation. Air enters a compressor at steady state at 1 bar, 290 K, 6 m/s through 0.1 m^2; exits at 7 bar, 450 K, 2 m/s. Heat loss is 180 kJ/min. Using the ideal gas model, find the power input (kW).

2017 Q2(c) 17 marks

A turbine-driven pump supplies water to a mixing chamber 25 m above the pump. Heat loss from the water is 2 kW; turbine heat transfer and PE negligible; KE negligible everywhere. Find (i) the pump power and (ii) the steam mass flow through the turbine. (Figure below.)



Full answer: ESE 1205 Study Pack, topic A6.

A3 Pure substance, phase rule, state postulate, T-v diagram, quality

Pass set · Very likely · 82% · Medium

Asked in 6 of 7 papers · 79 marks in total

STUDY THESE

- Pure substance = fixed chemical composition throughout (water + steam is; liquid air + gaseous air is not)
- Gibbs phase rule (non-reacting): $F = C - \pi + 2$
- State postulate: state of a simple compressible system is fixed by two independent intensive properties
- T-v diagram: compressed liquid, saturated liquid line, wet region, saturated vapour line, superheated region, critical point
- Quality $x = m_{vapour}/m_{total}$, $v = v_f + x\,v_{fg}$; steam table use (rigid tank cooled = constant v)

2023 Q1(a) 8 marks

Consider (i) a mixture of liquid and gaseous water and (ii) a mixture of liquid and gaseous air. Which one is a pure substance? Explain.

2023 Q1(b) 8 marks

What is state postulate? Write down Gibbs' phase rule for a non-reacting system and mention the meaning of each term.

2023 Q2(a) 10 marks

State the first law of thermodynamics. Draw the T-v diagram of a single-component pure substance and label constant pressure line, saturation line, critical point and regions.

2022 Q1(b) 4 marks

What is phase rule? Write down the phase rule for a non-reacting system.

2020 Q1(b) 6 marks

Write down the phase rule and mention the meaning of each term.

2020 Q1(c) 6 marks

Suppose (i) a mixture of liquid and gaseous water and (ii) a mixture of liquid and gaseous air. Which one is a pure substance? Explain.

2019 Q1(b) 8 marks

Define pure substance and phase. Discuss the condition based on which air can be treated as a pure substance.

2019 Q4(b) 5 marks

State the 'state postulate' of a simple compressible system. What is its significance?

2019 Q1(d) 10 marks

A closed, rigid tank filled with water, initially at 2 MPa, quality 80 % and volume 0.5 m^3, is cooled until the pressure is 0.4 MPa. Show the process on a sketch of the T-v diagram and evaluate the heat transfer.

2018 Q2(a) 7 marks

State the zeroth law of thermodynamics. Draw the T-v diagram of a single-component pure substance and label constant pressure line, saturation lines, critical point and constant quality lines.

2017 Q1(c) 7 marks

Write down the state postulate of a simple compressible system. Define quality and derive $x = (v - v_f)/(v_g - v_f)$.

Full answer: ESE 1205 Study Pack, topic A3.

A9 Exergy: dead state, closed-system and flow exergy, exergy balance, second-law efficiency

Next to add · Very likely · 100% · Hard

Asked in 7 of 7 papers · 131 marks in total

STUDY THESE

- Exergy = maximum useful work obtainable as the system comes to the dead state (T_0, p_0, rest, $z = 0$); exergy can only be destroyed, never generated
- Useful work $W_u = W - p_0(V_2 - V_1)$; $X_{destroyed} = T_0 S_{gen}$
- Closed system: $X = (U - U_0) + p_0(V - V_0) - T_0(S - S_0) + KE + PE$ (derivation)
- Flow exergy $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$; exergy balance for open system; $\eta_{II} = W_u/W_{rev}$

> 2023 Q4(c) 15 marks
>
> Show that the exergy of a flow stream is $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$.

> 2022 Q4(c) 4 marks
>
> What is exergy? Explain its physical significance.

> 2022 Q4(d) 18 marks
>
> A piston-cylinder holds 0.05 kg steam at 1 MPa, 300 °C. It expands to 200 kPa, 150 °C doing work, losing heat to surroundings at $T_0 = 25$ °C, $p_0 = 100$ kPa. Find (i) exergy at initial and final states, (ii) exergy change, (iii) exergy destroyed, (iv) second-law efficiency.

> 2021 Q4(a) 12 marks
>
> How does useful work differ from actual work? Define entropy and exergy. Show the closed-system exergy expression.

> 2021 Q4(b) 5 marks
>
> Write down the exergy balance equation for an open system and the meaning of each term.

> 2021 Q3(c) 20 marks
>
> Steam enters an adiabatic turbine at 6 MPa, 600 °C, 80 m/s and leaves at 50 kPa (printed "50 MPa"), 100 °C, 140 m/s. Power output 5 MW (printed "5 MPa"). Find (i)

mass flow, (ii) reversible power, (iii) isentropic efficiency, (iv) second-law efficiency. Surroundings 25 °C.

2020 Q3(b) 5 marks

What is dead state? Is it possible to generate exergy? If not, why?

2020 Q3(c) 15 marks

A 200 m^3 rigid tank holds compressed air at 1 MPa, 300 K. Find the work obtainable if the environment is at 100 kPa, 300 K.

2019 Q4(c) 15 marks

Define exergy. Prove that the maximum theoretical work of a system is $W_c = (U-U_0) + p_0(V-V_0) - T_0(S-S_0) + KE + PE$.

2018 Q4(a) 12 marks

What is exergy? Show that the exergy of a system is given by the same expression.

2017 Q4(b) 10 marks

What is exergy? Show that the exergy of a system is $\xi = (U-U_0) + p_0(V-V_0) - T_0(S-S_0) + KE + PE$.

Full answer: ESE 1205 Study Pack, topic A9.

A8 Entropy: Clausius inequality, Tds equations, entropy balance, isentropic efficiency

Pass set · Very likely · 100% · Hard

Asked in 7 of 7 papers · 127 marks in total

STUDY THESE

- Clausius inequality $\oint \delta Q/T \le 0$; definition $dS = (\delta Q/T)_{int\ rev}$
- Increase of entropy principle: $S_{gen} = \Delta S_{isolated} \ge 0$
- Tds equations: $T\,ds = du + p\,dv$, $T\,ds = dh - v\,dp$; ideal gas $s_2 - s_1 = c_v \ln(T_2/T_1) + R\ln(v_2/v_1) = c_p \ln(v_2/v_1) + c_v \ln(p_2/p_1)$; incompressible $ds = c\,dT/T$
- Entropy rate balance for a control volume; isentropic efficiency of turbine/compressor; steady-flow work $-\int v\,dp$

2023 Q3(c) 18 marks

Air is compressed adiabatically from 100 kPa, 12 °C to 800 kPa at 0.2 kg/s. Isentropic efficiency 80 %. Find (i) exit temperature, (ii) power input.

2022 Q3(d) 9 marks

Explain the increase of entropy principle.

2022 Q4(a) 9 marks

For an ideal gas derive $s_2 - s_1 = c_p \ln(v_2/v_1) + c_v \ln(p_2/p_1)$.

2022 Q3(c) 8 marks

Find the work to compress steam isentropically from 100 kPa to 1 MPa if the inlet is (i) saturated liquid, (ii) saturated vapour.

2021 Q3(b) 10 marks

State the Clausius inequality and prove $ds = (\delta Q/T)_{\text{int rev}}$. Write the entropy rate balance for a control volume and the meaning of each term.

2021 Q4(c) 5 marks

Derive the first and second Tds equations. (Real gas part in A4.)

2020 Q2(a) 6 marks

A system has only a work interaction with its surroundings. Does its entropy change? Explain.

2019 Q3(b) 10 marks

Derive the first and second Tds equations. Show that for an incompressible substance $s_2 - s_1 = \int c(T)dT/T$.

2018 Q3(b) 7 marks

Derive the first and second Tds equations. (KP statement in A7.)

2018 Q3(c) 9 marks

What is a reversible process? Show that $dS = (\delta Q/T)_{\text{int rev}}$.

2018 Q4(c) 7 marks

Write down the entropy rate balance equation for a control volume and describe each term.

2017 Q3(b) 11 marks

State the Clausius inequality and prove $dS = (\delta Q/T)_{\text{int rev}}$. Derive the first and second Tds equations.

2017 Q3(c) 6 marks

Write down the entropy rate balance equation for a control volume and the meaning of each term.

2017 Q3(d) 12 marks

Steam enters a turbine at 30 bar, 400 °C, 160 m/s and leaves as saturated vapour at 100 °C, 100 m/s. Work = 540 kJ/kg; heat transfer occurs at an outer surface temperature of 350 K. Find the entropy production per kg of steam.

Full answer: ESE 1205 Study Pack, topic A8.

A1 Basic concepts: system, viewpoints, properties, equilibrium, process

Pass set · Likely · 79% · Easy

Asked in 6 of 7 papers · 67 marks in total

STUDY THESE

- System, surroundings, boundary; closed (control mass), open (control volume), isolated, adiabatic
- Macroscopic (classical) vs microscopic (statistical) viewpoint; continuum
- Property, state, intensive vs extensive vs specific, internal vs external properties
- Thermodynamic equilibrium = mechanical + thermal + chemical (+ phase); quasi-equilibrium process; process, path, cycle; steady vs transient

`2023 Q1(c) 10 marks`

Write short notes on (i) quasi-equilibrium process, (ii) intensive property, (iii) thermodynamic equilibrium, (iv) thermodynamic system, (v) external property.

`2021 Q1(c) 5 marks`

Define continuum. Write down the Gibbs phase rule for a non-reacting system and mention the meaning of each term.

`2021 Q2(b) 5 marks`

What are the differences between steady state and transient analysis of a system? Explain with example.

`2020 Q1(a) 6 marks`

Define thermodynamic system and classify it.

`2019 Q1(a) 7 marks`

What does it mean by macroscopic and microscopic viewpoints of thermodynamics? Explain with example.

`2019 Q1(c) 10 marks`

Discuss thermodynamic equilibrium.

`2019 Q2(d) 6 marks`

> Write the definition (with figure if necessary): (i) open system, (ii) thermodynamic process, (iii) total energy of a system.

> 2018 Q1(a) 6 marks
> What does it mean by macroscopic and microscopic viewpoints of thermodynamics? Explain with example.

> 2018 Q1(c) 6 marks
> Define: (i) thermodynamic open system; (ii) pure substance and phase; (iii) Gibbs phase rule for non-reacting systems.

> 2017 Q1(a) 6 marks
> Define and write short notes on: (i) open system, (ii) thermodynamic equilibrium, (iii) quasi-equilibrium process.

Full answer: ESE 1205 Study Pack, topic A1.

A2 Energy: energy mind map, stored vs transient energy, heat vs work, path and point functions

Pass set · Likely · 75% · Easy

Asked in 6 of 7 papers · 74 marks in total

STUDY THESE

- Energy mind map: stored energy (macroscopic KE, PE; microscopic internal energy: sensible, latent, chemical, nuclear) and energy in transit (heat, work, mass flow)
- Heat = energy crossing a boundary because of a temperature difference; work = energy crossing for any other reason (moving boundary, shaft, electric)
- Heat and work are boundary phenomena and path functions (inexact differentials δQ, δW); properties are point functions (exact dV, dU)
- Same energy can be heat or work depending on where the boundary is drawn (electric oven example)

> 2022 Q2(a) 10 marks
> "An energy transfer can be heat or work, depending on system selection": explain the statement.

> 2021 Q1(a) 10 marks
> Why are heat and work path functions but thermodynamic properties point functions? Describe.

> 2021 Q1(b) 10 marks
>
> Briefly explain the stored and transient forms of energy with the help of a schematic diagram.

> 2020 Q1(d) 12 marks
>
> "An energy transfer can be heat or work, depending on system selection": explain the statement.

> 2019 Q2(c) 12 marks
>
> Draw an energy mind map. Define the different terms in the map.

> 2018 Q1(b) 10 marks
>
> Draw an energy mind map and describe each term in brief.

> 2017 Q1(b) 10 marks
>
> Draw an energy mind map and describe each term in it.

Full answer: ESE 1205 Study Pack, topic A2.

A4 Ideal gas and real gas: equations of state, compressibility factor

Next to add · Likely · 75% · Medium

Asked in 5 of 7 papers · 88 marks in total

STUDY THESE

- Ideal gas: $Pv = RT$, $PV = mRT = nR_uT$, $R = R_u/M$; when a real gas behaves ideally (low P, high T relative to critical)
- Compressibility factor $Z = Pv/RT$; principle of corresponding states (Z vs P_R, T_R)
- van der Waals: $(P + a/v^2)(v - b) = RT$, $a = 27R^2T_{cr}^2/64P_{cr}$, $b = RT_{cr}/8P_{cr}$ (derived from the critical-point conditions)
- Redlich-Kwong, Soave-Redlich-Kwong, virial equation; ideal vs real gas comparison
- Ideal gas process numericals: constant volume, constant temperature, constant pressure legs of a cycle

> 2023 Q1(d) 9 marks
>
> Determine the pressure exerted by 4.0 kg of CO in a 0.03 m^3 container at 220 K using the ideal gas law, van der Waals and Redlich-Kwong equations.

> 2022 Q1(c) 6 marks

Write short notes on (i) compressibility factor, (ii) principle of corresponding states.

`2022` `Q1(d)` `15 marks`

Determine the pressure of nitrogen at T = 175 K and v = 0.00375 m^3/kg using (i) the ideal gas equation of state and (ii) the Soave-Redlich-Kwong equation of state.

`2021` `Q1(d)` `10 marks`

0.45 kg of air in a piston-cylinder undergoes the same three-process cycle (1-2 constant v, 2-3 constant T expansion, 3-1 constant p compression). State 1: 300 K, 101.325 kPa; state 2: 200 kPa. (ii) Sketch on p-v, determine T_2 and (iii) v_3 in m^3/kg.

`2021` `Q4(c)` `3 marks`

When can a real gas be treated as an ideal gas? (The Tds part of this question is answered in A8.)

`2021` `Q4(d)` `10 marks`

Predict the pressure of nitrogen at T = 175 K and v = 0.00375 m^3/kg using (i) the ideal gas equation and (ii) the van der Waals equation. Compare with the experimental value 10 MPa. For N_2: P_{cr} = 3.39 MPa, T_{cr} = 126.2 K.

`2018` `Q2(d)` `5 marks`

What is compressibility factor? When a gas is modelled as an ideal gas, what are the equations to be used?

`2018` `Q4(b)` `6 marks`

Write down the virial equation of state. How is this equation used for substances?

`2017` `Q1(d)` `11 marks`

What are the equations used when a gas is modelled as an ideal gas? 1 kg of air in a piston-cylinder undergoes a cycle: 1-2 constant specific volume, 2-3 constant temperature expansion, 3-1 constant pressure compression. At state 1, T = 300 K, p = 1.012 bar; at state 2, p = 2.0 bar. (i) Sketch the cycle on p-v, (ii) find T_2, (iii) find v_3 (M = 29).

`2017` `Q4(c)` `8 marks`

State the van der Waals equation of state and calculate the unknown constants in terms of critical pressure and temperature.

`2017` `Q8(a)` `5 marks`

Write down the comparison between ideal gases and real gases.

Full answer: ESE 1205 Study Pack, topic A4.

A10 Thermodynamic relations: Maxwell relations, expansivity, compressibility, Joule-Thomson

Only if time · Possible · 46% · Hard

Asked in 2 of 7 papers · 43 marks in total

STUDY THESE

- Gibbs equations du = T ds − p dv, dh = T ds + v dp, da = −s dT − p dv, dg = −s dT + v dp → four Maxwell relations
- Why Maxwell relations: entropy cannot be measured; they replace entropy derivatives by measurable p-v-T derivatives
- Volume expansivity $\beta = (1/v)(\partial v/\partial T)_p$; isothermal compressibility $\kappa = -(1/v)(\partial v/\partial p)_T$; ideal gas $\beta = 1/T$, $\kappa = 1/p$
- Joule-Thomson coefficient $\mu = (\partial T/\partial p)_h$; inversion line and maximum inversion temperature

> 2023 Q4(a) 12 marks
>
> Explain the significance of Maxwell relations. How do they allow entropy changes to be found where direct measurement is impossible?

> 2023 Q4(b) 8 marks
>
> Short notes: (i) volume expansivity, (ii) isothermal compressibility.

> 2022 Q8(b) 13 marks
>
> What is a Maxwell relation? Why do we need Maxwell relations? Express them in equation form.

> 2022 Q8(a) 10 marks
>
> "A cooling effect cannot be achieved by throttling unless the fluid is below its maximum inversion temperature": explain.

Full answer: ESE 1205 Study Pack, topic A10.

Section B: Power cycles, refrigeration, gas mixtures, psychrometry

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B6 Refrigeration: vapour compression and vapour absorption systems

Next to add · Very likely · 100% · Easy

Asked in 7 of 7 papers · 82 marks in total

STUDY THESE

- Vapour compression (VCR): compressor → condenser → expansion valve → evaporator; p-h and T-s diagrams; COP = $(h_1-h_4)/(h_2-h_1)$
- Vapour absorption (VAR, ammonia-water): absorber + pump + generator replace the compressor; heat-operated
- VCR vs VAR comparison; advantages of VAR (low-grade heat, few moving parts, no lubrication problem)
- Ton of refrigeration, ice-plant load: $Q = m[c_w\Delta T + \text{latent heat}]$; power = $Q/COP/\eta_{motor}$

> 2023 Q7(a) 8 marks
> Differentiate between vapour compression and vapour absorption refrigeration systems.

> 2022 Q6(b) 12 marks
> With a neat schematic, describe the working principle of an ammonia-water absorption refrigeration system.

> 2021 Q7(a) 12 marks
> With a neat schematic, describe the working principle of a vapour compression refrigeration system. Write its advantages and disadvantages.

> 2020 Q5(a) 10 marks
> Write down the advantages of vapour absorption refrigeration over vapour compression refrigeration.

> 2019 Q7(a) 10 marks
> With a neat schematic, describe the working principle of a vapour absorption refrigeration system.

> 2018 Q6(b) 12 marks

> Why is VCR called work-operated and absorption refrigeration heat-operated? With the p-h diagram, describe the VCR mechanism.

> 2017 Q6(c) 8 marks
>
> Describe the mechanism of a vapour compression refrigeration system with p-h and T-s diagrams.

> 2017 Q6(d) 10 marks
>
> An ice plant produces 10 tonnes of ice per day at 0 °C from water at 20 °C. Estimate the compressor motor rating if COP = 2.5 and overall electro-mechanical efficiency = 90 %.

Full answer: ESE 1205 Study Pack, topic B6.

B1 Power cycle basics: air-standard assumptions, gas vs vapour cycles, Carnot cycle

Next to add · Very likely · 100% · Easy

Asked in 7 of 7 papers · 70 marks in total

STUDY THESE

- Classification of cycles: power vs refrigeration, gas vs vapour, closed vs open, internal vs external combustion
- Air-standard assumptions (air ideal gas, internally reversible, combustion → heat addition, exhaust → heat rejection); cold-air-standard adds constant c_p, c_v at 25 °C
- Carnot cycle: two isothermal + two isentropic processes; $\eta = 1 - T_L/T_H$; why the Carnot (vapour) cycle is not practical
- Mean temperature of heat addition: $\eta = 1 - T_0/T_m$; effect of boiler and condenser pressure

> 2023 Q5(a) 10 marks
>
> Write down the classification of thermodynamic cycles. Discuss the processes of the Carnot cycle with diagrams.

> 2022 Q6(a) 8 marks
>
> What is an air-standard cycle? What are its assumptions?

> 2021 Q5(a) 8 marks
>
> Write down the air-standard assumptions. What are the differences between vapour cycles and gas power cycles?

> 2020 Q4(a) 8 marks

> What is a power cycle? How does the Carnot efficiency vary with boiler and condenser pressure?

> 2020 Q4(b) 4 marks
> Why is the Carnot cycle not used in practice?

> 2019 Q6(a) 8 marks
> What is an air-standard cycle? Write down the air-standard assumptions.

> 2019 Q5(a) 8 marks
> What is mean temperature of heat addition? Show that cycle efficiency is a function of it.

> 2018 Q5(a) 7 marks
> What is an air-standard cycle? Write down the air-standard assumptions of an actual cycle.

> 2018 Q6(a) 5 marks
> Write down the basic differences between gas power cycles and vapour cycles. Why is the Carnot vapour power cycle not used in practice?

> 2017 Q5(c) 4 marks
> What is air-standard cycle efficiency? (Diesel derivation part in B4.)

Full answer: ESE 1205 Study Pack, topic B1.

B8 Psychrometry: moist-air properties, derivations, chart and numericals

Pass set · Very likely · 100% · Medium

Asked in 7 of 7 papers · 183 marks in total

STUDY THESE

- Dry air, moist air, saturated air; DBT, WBT, dew point; wet-bulb depression; why DBT ≥ WBT
- Specific humidity $\omega = 0.622\, p_v/(p - p_v)$; relative humidity $\phi = p_v/p_s$; degree of saturation $\mu = \omega/\omega_s$; $\phi = \mu/[1 - (1-\mu)p_s/p]$
- Carrier's equation $p_v = p_{wb} - (p - p_{wb})(t_{db} - t_{wb})/(1544 - 1.44\, t_{wb})$; $h = 1.005t + \omega(2500 + 1.88t)$
- Adiabatic saturation (constant enthalpy); psychrometric chart lines and processes (cooling and dehumidification); everyday questions (wet cloth, sweat)

2023 Q8(a) 10 marks

Short notes: (i) DBT, (ii) WBT, (iii) specific humidity, (iv) relative humidity.

2023 Q8(b) 10 marks

Derive the expression for degree of saturation.

2023 Q8(c) 5 marks

Draw and mention the different parameters of a psychrometric chart.

2022 Q8(c) 12 marks

Air at 1.0132 bar, DBT 32 °C, WBT 26 °C. Find (i) p_v, (ii) ϕ, (iii) ω, (iv) enthalpy.

2021 Q8(b) 10 marks

Short notes: (i) psychrometry, (ii) relative humidity, (iii) moist air, (iv) wet-bulb depression, (v) dew point temperature.

2021 Q8(a) 10 marks

With the psychrometric chart, describe the change of specific humidity and relative humidity during cooling and dehumidification.

2021 Q8(c) 15 marks

Air at 1.0132 bar, DBT 30 °C, WBT 27 °C. Find (i) p_v, (ii) ω, (iii) dew point, (iv) ϕ, (v) μ, (vi) density, (vii) enthalpy.

2020 Q6(b) 6 marks

What are absolute humidity, relative humidity and degree of saturation?

2019 Q8(b) 5 marks

Why does a wet cloth dry slower in summer than in winter?

2019 Q8(a) 5 marks

Why is the dry-bulb temperature always greater than the wet-bulb temperature?

2019 Q8(c) 9 marks

Prove that $\phi = \mu/[1 - (1 - \mu)(p_s/p)]$.

2019 Q8(d) 16 marks

Air at 1.0132 bar, DBT 35 °C, WBT 31 °C. Find (i) p_v, (ii) ω, (iii) ϕ, (iv) dew point, (v) enthalpy, (vi) specific volume.

2018 Q8(b) 5 marks

Why do we feel sweaty in summer and dry in winter?

2018 Q8(a) 8 marks

Define (i) psychrometrics, (ii) relative humidity, (iii) degree of saturation, (iv) dew point temperature.

> **2018 Q8(c) 10 marks**
>
> Show that the enthalpy of the mixture remains constant during an adiabatic saturation process.

> **2018 Q8(d) 12 marks**
>
> Air at 1.0132 bar, DBT 32 °C, WBT 24 °C. Find (i) p_v, (ii) ω, (iii) ϕ, (iv) μ, (v) enthalpy.

> **2017 Q7(a) 7 marks**
>
> What is meant by ‘psychrometry’? Why does a wet cloth dry faster in winter than in summer?

> **2017 Q7(b) 8 marks**
>
> Distinguish between dry-bulb and wet-bulb temperature. What is degree of saturation?

> **2017 Q7(c) 10 marks**
>
> Derive an expression for specific humidity and show that it is a function of vapour pressure and barometric pressure.

> **2017 Q7(d) 10 marks**
>
> Air at DBT 30 °C, WBT 18 °C, barometer 760 mm Hg. Find (i) p_v, (ii) ϕ, (iii) dew point.

Full answer: ESE 1205 Study Pack, topic B8.

B7 Ideal gas mixtures: Dalton, Amagat, mole and mass fractions, mixture properties

Pass set · Very likely · 100% · Medium

Asked in 7 of 7 papers · 172 marks in total

STUDY THESE

- Gravimetric (mass) and volumetric/molar analysis; mass fraction $mf_i = m_i/m$, mole fraction $y_i = n_i/n$; $\Sigma = 1$
- Dalton: $p = \Sigma p_i$, $p_i = y_i p$; Amagat: $V = \Sigma V_i$, $V_i = y_i V$; for ideal gases volume fraction = mole fraction
- Apparent molar mass $M_m = \Sigma y_i M_i$; $R_m = R_u/M_m$; $c_{p,m} = \Sigma mf_i c_{p,i}$; $c_{p,i} = \gamma R_i/(\gamma-1)$
- Numerical recipe: $n_i = m_i/M_i \rightarrow y_i \rightarrow M_m \rightarrow R_m \rightarrow V = mR_mT/p \rightarrow \rho \rightarrow c_p, c_v$

> **2023 Q8(d) 10 marks**

3 kg N_2 + 5 kg CO_2 at 300 kPa, 20 °C. Find (i) mole fractions, (ii) equivalent molecular weight.

2022 Q7(b) 9 marks

State Dalton's law. What is meant by gravimetric and volumetric analysis?

2022 Q7(c) 16 marks

5 kg N_2 + 6 kg CO_2 at 4 bar, 27 °C. Find (i) mole fractions, (ii) molecular weight, (iii) gas constant, (iv) volume and density.

2021 Q7(b) 4 marks

State Dalton's law of partial pressure and Amagat's law.

2021 Q7(d) 6 marks

Short notes: (i) gas mixture, (ii) gravimetric analysis, (iii) partial pressure of a gas mixture.

2021 Q7(c) 13 marks

Mixture of 60 % O_2 and 40 % CO_2 (mole basis) at 300 kPa, 20 °C. Find (i) mass fractions, (ii) molecular weight, (iii) gas constant, (iv) partial pressures and partial volumes, (v) volume, density, c_p, c_v (per kmol of mixture).

2020 Q6(a) 6 marks

State Dalton's law of partial pressure and Amagat's law.

2020 Q6(c) 18 marks

3 kg N_2 + 5 kg CO_2 at 300 kPa, 20 °C. Find (i) mole fractions, (ii) molecular weight, (iii) gas constant, (iv) partial pressures, (v) density, (vi) c_p, c_v.

2019 Q7(b) 4 marks

Define: (i) gas mixture, (ii) gravimetric analysis, (iii) volumetric analysis, (iv) ideal gas.

2019 Q7(c) 6 marks

State Dalton's law of partial pressure. Show that partial pressure = mole fraction × total pressure.

2019 Q7(d) 15 marks

3.5 kg N_2 + 5.5 kg CO_2 at 300 kPa, 200 °C. Find (i) mass and mole fractions, (ii) molecular weight, (iii) volume and density, (iv) c_p, c_v (γ CO_2 = 1.29, N_2 = 1.40).

2018 Q7(a) 10 marks

State Dalton's law. Show that $M_m = \Sigma x_i M_i$.

> **2018 Q7(b) 15 marks**
>
> A 300 L drum holds a gas mixture at 600 kPa, 40 °C: 35 % O_2, 65 % CH_4 by volume. To get 70 % O_2, 30 % CH_4 at the same p and T, how many kg of mixture must be bled off and what mass of O_2 added?

> **2018 Q7(c) 10 marks**
>
> A mixture of CH_4, N_2, CO and O_2 has partial pressures 140, 55, 70 and 15 kPa. Find the volumetric and gravimetric analysis.

> **2017 Q8(b) 8 marks**
>
> What is the apparent molar mass of a gas mixture? Does every molecule have this mass?

> **2017 Q8(c) 12 marks**
>
> For a homogeneous ideal gas mixture at T, p, V: (i) derive Dalton's law, (ii) show Σ mole fractions = 1, (iii) find the gas constant of the mixture.

> **2017 Q8(d) 10 marks**
>
> A mixture of 3 kg O_2, 5 kg N_2 and 12 kg CH_4. Find (i) mole fractions, (ii) average molar mass, (iii) gas constant.

Full answer: ESE 1205 Study Pack, topic B7.

B4 Otto, Diesel and Dual cycles (IC engines): diagrams, derivations, comparisons, numericals

Pass set · Very likely · 100% · Medium

Asked in 7 of 7 papers · 149 marks in total

STUDY THESE

- Engine terms: TDC, BDC, stroke, clearance volume, compression ratio r, cut-off ratio r_c, pressure ratio r_p, MEP; four-stroke SI engine
- Otto (constant-volume heat addition): $\eta = 1 - 1/r^{\gamma-1}$; Diesel (constant pressure): $\eta = 1 - (1/r^{\gamma-1})(r_c^{\gamma}-1)/[\gamma(r_c-1)]$
- Dual (limited pressure): heat added partly at constant v, partly at constant p; $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$ for same r and q_{in}; reverse for the same p_{max}
- Numericals: $T_2 = T_1 r^{\gamma-1}$, $q = c_v\Delta T$ or $c_p\Delta T$, $MEP = w_{net}/(v_1-v_2)$

> **2023 Q7(b) 12 marks**
>
> Describe the working principle of a four-stroke Otto cycle engine with diagrams.

2023 Q7(c) 15 marks

Air-standard Diesel: r = 16, 15 °C, 0.1 MPa, maximum temperature 1480 °C. Find (i) cut-off ratio, (ii) heat supplied per kg, (iii) efficiency, (iv) MEP.

2022 Q7(a) 10 marks

Do diesel or petrol engines operate at higher compression ratios? Why?

2021 Q5(b) 15 marks

With p-v and T-s diagrams, show that $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$ for the same compression ratio and heat rejection. Why?

2020 Q4(c) 18 marks

Air-standard Diesel cycle: r = 17, start of compression 20 °C, 100 kPa; temperature at end of constant-pressure heating 1500 °C. Find (i) cut-off ratio, (ii) heat supplied per kg, (iii) efficiency.

2019 Q6(b) 7 marks

From the derivation, show that the Diesel cycle efficiency is less than the Otto cycle efficiency for the same compression ratio.

2019 Q6(c) 20 marks

Air-standard Dual cycle: start of compression 0.1 MPa, 27 °C; end of compression 2.3 MPa; 430 kJ/kg supplied at constant volume; pressure 0.3 MPa at end of isentropic expansion. Find p and T at all points and the efficiency.

2018 Q5(b) 10 marks

Why is the dual cycle called a limited pressure cycle? Show from T-s diagrams and derivations that its efficiency lies between Otto and Diesel.

2018 Q5(c) 18 marks

Limited pressure cycle: r = 18, compression begins at 0.1 MPa, 40 °C; p_{max} = 7 MPa; heat added 1.675 MJ/kg. Find (i) heat at constant volume, (ii) heat at constant pressure, (iii) work, (iv) efficiency, (v) temperature after constant-volume heating, (vi) cut-off ratio.

2017 Q5(a) 6 marks

Draw the Diesel, Brayton and Otto cycles on p-v and T-s planes showing heat and work transfer.

2017 Q5(c) 8 marks

Derive an expression for the air-standard efficiency of the Diesel cycle in terms of cut-off ("pressure") ratio and compression ratio.

2017 Q5(d) 10 marks

> In an Otto cycle the temperatures at the beginning and end of isentropic compression are 316 K and 596 K. Find the air-standard efficiency and compression ratio ($\gamma = 1.4$).

Full answer: ESE 1205 Study Pack, topic B4.

B3 Rankine cycle numericals (simple, actual, reheat, regenerative)

Pass set · Very likely · 96% · Hard

Asked in 6 of 7 papers · 100 marks in total

STUDY THESE

- Recipe: state 1 sat. liquid at p_{cond}; $w_p = v_1(p_2 - p_1)$; state 3 from superheat table; $s_4 = s_3 \rightarrow x_4 \rightarrow h_4$
- Actual: $h_4 = h_3 - \eta_T(h_3 - h_{4s})$, $w_p = w_{p,s}/\eta_P$
- Reheat with a quality limit: $x_6 \rightarrow s_6 = s_5 \rightarrow$ find p_{reheat} where $s(p, 600\ °C) = s_6$
- Open FWH: $y = (h_3 - h_2)/(h_6 - h_2)$; $q_{out} = (1-y)(h_7 - h_1)$

> `2023` `Q5(c)` `13 marks`
>
> Same reheat problem (15 MPa, 600 °C, 10 kPa, moisture ≤ 10.4 %): reheat pressure and efficiency.

> `2022` `Q5(c)` `18 marks`
>
> Ideal regenerative cycle (printed "refrigerative") with one open FWH: 15 MPa, 600 °C, 10 kPa, bleed at 1.2 MPa. Find the bleed fraction and efficiency.

> `2021` `Q6(c)` `15 marks`
>
> Ideal reheat Rankine: HP turbine inlet 15 MPa, 600 °C; condenser 10 kPa; LP exit moisture ≤ 10.4 %; reheated to 600 °C. Find (i) reheat pressure, (ii) efficiency.

> `2020` `Q5(b)` `20 marks`
>
> Ideal Rankine cycle: turbine inlet 3 MPa, 355 °C, condenser 11 kPa. Find the thermal efficiency (steam table extract given in the paper).

> `2019` `Q5(d)` `16 marks`
>
> Rankine cycle: turbine inlet 4 MPa, 400 °C; condenser at 27 °C. Find the thermal efficiency; then with turbine $\eta = 85$ % and pump $\eta = 87$ %.

> `2018` `Q6(c)` `18 marks`
>
> Ideal regenerative cycle with one open FWH: 15 MPa, 550 °C, condenser 10 kPa, bleed at 1.2 MPa. Find the bleed fraction and efficiency.

Full answer: ESE 1205 Study Pack, topic B3.

B2 Rankine cycle theory: ideal vs actual, efficiency improvement, superheat, reheat, regeneration

Pass set · Likely · 79% · Medium

Asked in 5 of 7 papers · 95 marks in total

STUDY THESE

- Ideal Rankine: 1-2 pump (isentropic), 2-3 boiler (constant p), 3-4 turbine (isentropic), 4-1 condenser; $\eta = (w_t - w_p)/q_{in}$
- Actual cycle: friction pressure drops, heat loss, turbine and pump irreversibility (η_T, η_P)
- Raise efficiency: lower condenser pressure, superheat, raise boiler pressure (each raises $T_{m,in}$ or lowers T_{out}); superheat reduces moisture
- Reheat (keeps exit quality high at high boiler pressure; optimum reheat pressure ≈ 1/4-1/5 of boiler pressure); regeneration with open and closed feedwater heaters; cogeneration

> 2023 Q5(b) 12 marks
>
> Describe the methods of increasing the efficiency of the Rankine cycle with diagrams.

> 2023 Q6(a) 12 marks
>
> Derive the expression for the efficiency of an ideal regenerative Rankine cycle.

> 2023 Q6(b) 8 marks
>
> Differentiate between open and closed feedwater heaters.

> 2022 Q5(a) 10 marks
>
> How can we increase the efficiency of the Rankine cycle?

> 2022 Q5(b) 7 marks
>
> Why is regeneration done in the Rankine cycle? What is cogeneration?

> 2021 Q6(a) 10 marks
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> Why is steam superheated in the Rankine cycle? Why is the ideal cycle modified to the reheat cycle?

> 2021 Q6(b) 10 marks

With diagrams, describe the regenerative Rankine cycle with an open feedwater heater.

2019 Q5(b) 6 marks

From the T-s diagram, show the deviation of an actual Rankine cycle from the ideal one.

2019 Q5(c) 5 marks

Why is the ideal Rankine cycle modified to the reheat and regenerative Rankine cycles?

2017 Q6(a) 7 marks

Describe the Rankine cycle using T-s and h-s diagrams and give its thermal efficiency.

2017 Q6(b) 8 marks

When is reheating recommended in a steam power plant? How is the reheat pressure optimised?

Full answer: ESE 1205 Study Pack, topic B2.

B5 Brayton (gas turbine) cycle

Next to add · Likely · 64% · Medium

Asked in 3 of 7 papers · 42 marks in total

STUDY THESE

- Processes: 1-2 isentropic compression, 2-3 constant-p heat addition, 3-4 isentropic expansion, 4-1 constant-p heat rejection
- $\eta = 1 - 1/r_p^{(\gamma-1)/\gamma}$; $T_2/T_1 = r_p^{(\gamma-1)/\gamma}$
- Back work ratio = w_c/w_t; with component efficiencies $T_{2a} = T_1 + (T_{2s}-T_1)/\eta_C$, $T_{4a} = T_3 - \eta_T(T_3-T_{4s})$

2023 Q6(c) 15 marks

Ideal Brayton cycle: pressure ratio 8, 300 K and 1300 K. Using air-standard assumptions find (i) compressor and turbine exit temperatures, (ii) efficiency.

2022 Q6(c) 15 marks

Air-standard Brayton cycle between 300 K and 1200 K and pressure limits 101 kPa and 505 kPa. Find (i) efficiency, (ii) compressor work, (iii) turbine work (kJ/kg).

2021 Q5(c) 12 marks

> Brayton cycle: pressure ratio 8, 300 K compressor inlet, 1300 K turbine inlet; η_C = 80 %, η_T = 85 %. Find (i) exit temperatures of compressor and turbine, (ii) back work ratio, (iii) thermal efficiency.

Full answer: ESE 1205 Study Pack, topic B5.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
BDC	bottom dead centre
CI	compression-ignition
CM	control mass
COP	coefficient of performance
CV	control volume
DBT	dry-bulb temperature
DPT	dew-point temperature
EOS	equation of state
FWH	feedwater heater
HP	high-pressure
IAPWS	International Association for the Properties of Water and Steam
IC	internal combustion
KE	kinetic energy
kmol	kilomole (molar mass in kilograms)
KP	Kelvin-Planck (statement)
kPa	kilopascal
kW	kilowatt
LP	low-pressure
MEP	mean effective pressure
MPa	megapascal

MW	megawatt
PE	potential energy
PMM1	perpetual-motion machine of the first kind
PMM2	perpetual-motion machine of the second kind
RH	relative humidity
RK	Redlich-Kwong
SFEE	steady-flow energy equation
SI	spark-ignition
SRK	Soave-Redlich-Kwong
TDC	top dead centre
Tk	Bangladeshi taka
TR	ton of refrigeration (3.5 kW)
VAR	vapour absorption refrigeration
VCR	vapour compression refrigeration
vdW	van der Waals
WBT	wet-bulb temperature

Built from the scanned ESE 1205 papers (2017, 2018, 2019, 2020, 2021, 2022, 2023). Question text is cleaned up from the scans.